Group -28 Assignment - 5

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4-bit Adder/Subtractor using 74LS83 and 74LS86

Aim

To design and implement a 4-bit Adder/Subtractor circuit using TTL IC 74LS83 (4-bit full adder) and 74LS86 (XOR gates). Inputs are applied using switches and outputs are observed using LEDs. The operation is verified for given 4-bit 2's complement inputs.

Apparatus Required

- Breadboard
- IC 74LS83 (4-bit full adder)
- IC 74LS86 (Quad XOR)
- Switches for inputs A3–A0, B3–B0 and MODE (Cin)
- LEDs for outputs S4–S1 and Cout
- Connecting wires and +5V DC supply

Theory

The 74LS83 is a 4-bit full adder which performs:

$$S = A + B + C_{in}$$

Subtraction is performed using 2's complement:

$$A - B = A + (2's\ complement\ of\ B)$$

$$2's\ complement\ of\ B = \overline{B} + 1$$

So,

$$A - B = A + \overline{B} + 1$$

Using XOR gates for conditional inversion:

$$B_i' = B_i \oplus C_{in}$$

and connect:

$$C_{in} = \text{MODE}$$

Hence the adder performs:

$$S = A + (B \oplus MODE) + MODE$$

- MODE = 0 $\Rightarrow$ Addition: $A + B$
- MODE = 1 $\Rightarrow$ Subtraction: $A - B$

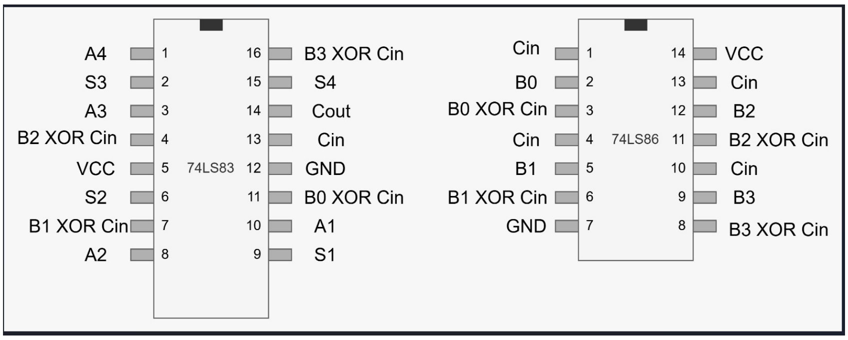

Figure 1: circuit diagram.

Pin Connections

74LS83 Pin Connections

Power:

- Pin 5 → +5V (VCC)
- Pin 12 → GND

A Inputs (from switches):

- Pin 10 → A1 (LSB)
- Pin 8 → A2
- Pin 3 → A3
- Pin 1 → A4 (MSB)

B Inputs (from XOR outputs of 74LS86):

- Pin 11 → (B0 XOR Cin)
- Pin 7 → (B1 XOR Cin)
- Pin 4 → (B2 XOR Cin)
- Pin 16 → (B3 XOR Cin)

Carry Input:

- Pin 13 → Cin (MODE switch)

Sum Outputs (to LEDs):

- Pin 9 → S1 (LSB) LED
- Pin 6 → S2 LED
- Pin 2 → S3 LED
- Pin 15 → S4 (MSB) LED

Carry Output:

- Pin 14 → Cout LED

74LS86 Pin Connections

Power:

- Pin 14 → +5V (VCC)
- Pin 7 → GND

Cin / MODE distribution:

- Cin is connected to pins 1, 4, 10, 13 of 74LS86

B Inputs (from switches):

- Pin 2 → B0
- Pin 5 → B1
- Pin 12 → B2
- Pin 9 → B3

XOR Outputs (to 74LS83 B inputs):

- Pin 3 → (B0 XOR Cin) → 74LS83 Pin 11
- Pin 6 → (B1 XOR Cin) → 74LS83 Pin 7
- Pin 11 → (B2 XOR Cin) → 74LS83 Pin 4
- Pin 8 → (B3 XOR Cin) → 74LS83 Pin 16

Final Summary of Working

- If Cin=0, XOR outputs equal B, hence circuit performs addition.
- If Cin=1, XOR outputs become $\overline{B}$ and Cin adds +1, hence circuit performs subtraction.

Procedure

1. Insert IC 74LS83 and IC 74LS86 on the breadboard
2. Connect power pins: 74LS83 (VCC=5, GND=12) and 74LS86 (VCC=14, GND=7).
3. Connect A inputs from switches to 74LS83 pins 10, 8, 3, 1.
4. Connect B inputs from switches to 74LS86 pins 2, 5, 12, 9.
5. Connect Cin (MODE) switch to:
   - 74LS83 pin 13
   - 74LS86 pins 1, 4, 10, 13
6. Connect XOR outputs to 74LS83 B inputs:
   - 74LS86 pin 3 → 74LS83 pin 11
   - 74LS86 pin 6 → 74LS83 pin 7
   - 74LS86 pin 11 → 74LS83 pin 4
   - 74LS86 pin 8 → 74LS83 pin 16
7. Connect outputs S1–S4 and Cout to LEDs through resistors.
8. Set Cin=0 and verify addition.
9. Set Cin=1 and verify subtraction.
10. Apply the test inputs and record the outputs.

Observation Table (4-bit 2's complement)

Case	Operation	Cin	A(dec)	A(bin)	B(dec)	B(bin)	S(bin)	Cout	S(dec)
i	3 + 4	0	3	0011	4	0100	0111	0	7
ii	6 - 7	1	6	0110	7	0111	1111	0	-1
iii	(-3) + (-5)	0	-3	1101	-5	1011	1000	1	-8
iv	(-8) - (-2)	1	-8	1000	-2	1110	1010	0	-6

Result

A 4-bit Adder/Subtractor circuit was successfully designed using IC 74LS83 and 74LS86. The circuit correctly performed addition and subtraction for the given 4-bit 2's complement test cases.